

Challenge (Habanero)

- Q1.** A new gaming platform has 2 : 3 ratio of free to paid games. If there are 90 free games, how many paid games are there?
(A) 45
(B) 60
(C) 90
(D) 120
(E) 135
- Q2.** In a coding competition, 3 : 5 ratio of participants used Python to Java. If 30 participants used Java, how many used Python?
(A) 10
(B) 15
(C) 18
(D) 20
(E) 25
- Q3.** A gamer scored 4 : 7 ratio of wins to losses. If the gamer had 28 losses, how many wins did the gamer have?
(A) 10
(B) 12
(C) 16
(D) 20
(E) 24
- Q4.** A data plan has a 1 : 4 ratio of free data to paid data. If 12 GB is paid data, how much free data is included?
(A) 1 GB
(B) 2 GB
(C) 3 GB
(D) 4 GB
(E) 6 GB
- Q5.** To develop an app, the time spent on coding to testing is in 2 : 3 ratio. If the total time spent is 50 hours, how many hours are spent on coding?
(A) 15
(B) 18
(C) 20
(D) 25
(E) 30
- Q6.** The ratio of players playing on PC to console is 5 : 4. If there are 25 console players, how many PC players are there?
(A) 20
(B) 25
(C) 28
(D) 30
(E) 31.25
- Q7.** A programmer types 4 : 5 ratio of code to comments. If the programmer writes 20 lines of comments, how many lines of code does the programmer write?
(A) 12
(B) 15
(C) 16
(D) 20
(E) 25
- Q8.** A gaming tournament has 3 : 2 ratio of teams from the US to teams from Europe. If there are 18 teams from Europe, how many teams are from the US?
(A) 12
(B) 18
(C) 21
(D) 24
(E) 27

Open Ended Questions (FRQ)

- Q9.** A company has 3 servers with a ratio of 2 : 5 of gaming servers to database servers. If they add 2 more servers, what should be the new ratio of gaming to database servers to maintain the original proportion? Show your work and explain.
- Q10.** A coder is working on a project that requires 15 hours of coding and 10 hours of testing. If the coder works 4 hours a day, what ratio of days should be spent on coding to testing to complete the project on time? Use $\frac{a}{b}$ format for your ratio answer and provide a brief explanation.

Chef's Tips

- Ensure students use cross-multiplication for solving proportions
- Emphasize the importance of maintaining the original proportion when adding or removing quantities
- Remind students to check their units and ensure they are consistent throughout their calculations